Visualization of Text Data Visualization and the Challenges of Handling Text Data Transcript

Hello everyone, welcome to BITS Pilani Digital. Today, we will continue our learning journey for the course Data Visualisation and Storytelling. We will be continuing our module Designing Visuals, Visual Best Practises.

In today's class, we will be dealing with a very interesting topic of how to visualise textual data. In the last class, we have dealt with structural data and handling of the structural data and visualising the same. Today, we will be understanding how to visualise unstructured data, specifically the text data.

You would recall, structured data pertains to data being arranged in a well-defined format in the form of matrices, whereas unstructured data, the very name suggests there is no structure to follow. It becomes computationally very challenging to analyse the unstructured data. Resultantly, it becomes even more challenging to visualise the text data or the unstructured data.

Where is this unstructured data coming from? The data that is not structured is unstructured, primarily in the form of text messages, emails, articles that you encounter in your day-to-day life, where there is a lot of information exchanges happening. For example, we humans are writing, communicating through a written material for a long period of time, and though visual perception is very much preferred, we also communicate verbally in the text format. And most of the time, some crucial information, behavioural tendencies, and a lot of interesting insights are in the data patterns that are contained in the text language.

It is important for us to understand how to draw insights from such information contained in the textual messages and also do a similar analysis like that of the structured data. So now, we will move to understanding the unstructured data and address the challenges in analysing the textual data and visualising the textual data. However, to analyse textual data, you need the help of certain programming languages.

Textual data need to be first converted into a format that is computer-friendly or computational-friendly. For this, you need algorithms which are available in Python or Power BI or Tableau packages, which you can take the help and try to present the textual data in a visual format. So first of all, we will try to understand what is the challenges in textual data.

As I was mentioning, whenever you come across a text, you cannot immediately decipher the meaning from it unless you read it. However, you can convert it into an interpretable form by following certain steps.

This is what we call the pre-processing step of the textual data.

And you will be learning about these techniques in other courses. However, we will now briefly touch upon the pre-processing pipeline and understand how the textual data, let us say in the form of an email or a Twitter message or a news article is transformed into a form that is computational-friendly, which we will use for doing the visualisations. We will now focus only on the visualisation part, but giving you a brief understanding of the pre-processing pipeline.

First and foremost, when you get a text data, you have to convert that into same case. All the words have to either be in uppercase or in lowercase. Then when you get to the numbers which are there in the text, you have to either remove them or make them into words.

For example, there is 120 number, numerical 120, you either choose to remove it, but if you want to retain the meaning, you can convert it into 120 into the text. Then you have to remove the punctuation and special characters. This is the basic cleaning that you have to do to the data.

First convert the case, remove or adjust the numbers, then remove the special characters. This is the basic cleaning that you do for your textual data. Then comes the step called tokenization, where your text is broken into small pieces or converted into a bag of words.

So it is broken into very small pieces. For example, every limit for the visualisation of text data would be broken into the visualisation of text data into four parts. This is shown to you as a bag of words.

Now then you move to the part called stop words, removing the stop words, because certain words will not add much value to your text. For example, an apple versus apple will carry the same meaning. So you would remove certain stop words like a and d and so your text becomes less heavy to handle.

Generally, text data is large and voluminous. So you try to reduce unnecessary computational burden by removing the stop words. Then you do the stemming, which is trying to cut words which are common or make words which are into common parts.

For example, stemming is retaining the common portion of the words. For example, there is enjoyed and enjoyable. The word enjoy is there in the both the words and tries to convey the same meaning.

Therefore, you try to pick the word enjoy, which will convey the meaning in both the context. However, caution is also there because consider the word happiness and happily. In both the case, you have happy H H E P. So by taking that as a common word, you cannot get the meaning.

You have to be careful how stemming is done. Therefore, there is one more layer called lemmatisation, which is you are trying to get the root words of the words together or transform the words into root words. For example, increased increasingly increasing can be taught as a single word root word called increase.

They show the function of something increasing. So you can just transform them into a word called increase. So this preprocessing has to be done before you do any analysis on the textual data.

Thankfully, you have algorithms which will do that for you and you can directly look at the visualisations. You will be taught about these aspects more detailedly in some other courses

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